

Charlie Hoose

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WORK & RESEARCH EXPERIENCE

First Citizens Bank

Pasadena, CA

Business Analyst I

July 2026 – Present

- Reduce ongoing team data-request processing time by 5 hours/week across 90+ requests through an automated Share-Point intake system with Power Automate workflows and a companion Power BI approval/status-tracking dashboard
- Improve monthly KPI reporting accuracy by 10% across 8 community development reports for 30+ stakeholders, including senior bank leadership, through pivot-table and lookup-based variance checks that flag discrepancies
- Cut quarterly FDIC compliance reporting turnaround time by 70% by using text-search and index/match functions to convert raw data into structured dummy variables, then summarizing results via pivot tables

Data Analyst Intern

June 2025 – June 2026

- Leveraged SQL and Netezza to enhance philanthropy dashboards, joining new data fields to improve reporting accuracy
- Engineered multiple Excel VBA macros to automate dashboard color-coding for monthly reporting, reducing a 45-minute manual process to under 10 seconds with 100% accuracy for 5+ reports
- Streamlined Low-to-Moderate Income rate analysis by executing complex data joins and transformations in Excel using RegEx, saving \$250,000 in loan payments
- Developed an automated ETL reconciliation workflow in Excel to detect mismatching payments across tracking systems, saving 93% of manual processing time with 97% accuracy
- Led an intern team to analyze historic and forecasted philanthropy donations, creating and presenting Tableau dashboards to 60+ employees and informing a bank-wide philanthropic strategy

Italy National Baseball Team

Los Angeles, CA

Data Scientist

January 2026 – March 2026

- Trained an XGBoost model predicting pitcher-batter matchup outcomes on 2020–2024 MLB pitch-level data, achieving an R^2 of 0.40 and OPS bucketed correlation of 0.974 on 2025 holdout data
- Built an R Shiny app with player/team presets generating exploitability grids, delivering 7+ predictive dashboards that directly informed World Baseball Classic batting lineup and pitching strategy for the Italian National Team

B. John Garrick Institute for the Risk Sciences at UCLA

Los Angeles, CA

Data Science Research Intern

April 2024 – July 2025

- Designed an NLP pipeline to parse wildfire research papers, extracting keywords and contextual data to enhance evacuation model feature importance and cutting research synthesis time by over 80%
- Analyzed 5 competing wildfire evacuation platforms to identify performance gaps, presenting findings via matplotlib and seaborn visualizations to inform our team's modeling priorities
- Centralized findings across human and traffic evacuation model research into 3 reports, creating a referenceable knowledge base that informed I-Corps presentations

PROJECTS

FinQuery: Language to SQL Output Project

June 2026

- Enabled non-technical users to self-serve financial insights by developing a full-stack Streamlit application that translates natural language questions into SQL queries and auto-generates corresponding visualizations
- Validated reliability with 100% accuracy on a 25-query adversarial test set spanning 5 relational tables, stress-testing the Claude API pipeline against ambiguous and edge-case financial questions

Wildfire Acreage Prediction Project

April 2026

- Interviewed wildfire and GIS experts to source and validate national fire response datasets for more accurate modeling
- Performed data cleaning in pandas and ggplot2 visualizations to identify key trends influencing wildfire spread
- Constructed a wildfire size classification model using XGBoost, achieving 83% accuracy and 0.82 weighted F1 score
- Containerized wildfire prediction model with Docker and deployed as a live REST API endpoint on Azure App Service

EDUCATION

University of California, Los Angeles (UCLA)

June 2026

B.S. in Statistics and Data Science, Minor in Data Science Engineering

GPA: 3.92/4.00

- **Relevant Coursework:** Data Structures and Algorithms, Machine Learning, Deep Learning & Neural Networks, Statistical Modeling and Data Mining, Data Analysis and Regression, Relational Databases, Time Series Analysis
- **Club Involvement:** Data Science Union (Data Scientist, Dir. of Membership, Executive Advisor), Statistics Club
- **LinkedIn Learning Certifications:** Financial Planning & Analysis (FP&A) Foundations, Excel for FP&A, Corporate Financial Statement Analysis, Leveraging Generative AI in Finance and Accounting

SKILLS

- **Languages:** Python, R, SQL, C++
- **Libraries/Frameworks:** Pandas, NumPy, Matplotlib, seaborn, scikit-learn, LangChain, ggplot2, tidyverse, dplyr
- **Tools:** Tableau, Excel (VBA, Pivot Tables, Macros, Power Query), PowerBI, Git/GitHub, LaTeX, Kadince, R Markdown, Power Automate, Azure (Azure App Service, Azure ML, Azure CLI), Flask, Docker, SQLite, Claude Code
- **Methods:** Feature Engineering, ETL, Financial Analysis, Regression, Classification, Clustering, Statistical Inference